J. ROBERT MANUEL

Organist · Pianist · Choir Director · Composer · Teacher | Software Quality Assurance (SQA) & V&V Professional

144 Jarod Drive, Moon Township, PA 15108 · 724-272-1175 · pianomanuel@gmail.com · LinkedIn

PROFESSIONAL SUMMARY

Organist, pianist, choir director, composer, and teacher whose primary calling is sacred music — including fifteen years as full-time Director of Music Ministries at St. Stephen's Episcopal Church, Sewickley — with a parallel career since 2011 in FDA-regulated software quality assurance as a Lead V&V and Lead SQA (Level IV) engineer on medical-device, clinical-trial, healthcare, and embedded / IoT systems. The through-line is craft under pressure: translating the end user's day-in-the-life into work that matches the field.

CORE COMPETENCIES

Music, Worship & Teaching

- Pipe organ and piano performance; choir direction for adult, men-and-boys, and university choirs; SATB rehearsal technique
- Liturgical music direction in the Episcopal tradition (Book of Common Prayer 1979); contemporary worship-team leadership
- Sacred choral composition and engraving (SATB and organ); arranging and program-building for touring ensembles
- Collegiate teaching: ear training, music theory, organ studio, group keyboard; mentoring of worship leaders and young musicians

Software Quality Assurance & Regulated Systems

- Verification & Validation (V&V) of FDA Class I and Class II medical devices and control software; full-SDLC SQA
- Regulatory frameworks: FDA QSR 21 CFR Part 820, ISO 13485, IEC 62304, ISO 14971, IEC 60601, ISO 9001; HIPAA
- Requirements analysis and traceability; test planning, protocol authoring and execution, dry runs, verification reports
- Functional, negative, stress, integration, UI, performance, and embedded / IoT testing; DTR / defect capture and resolution
- VMware ESXi virtual test labs and VM templates for unattended automated testing; back-end verification with SQL
- QA team leadership, SQA mentoring, and developer–QA "translation" in Agile / daily-scrum environments

PROFESSIONAL EXPERIENCE — MUSIC MINISTRY & TEACHING

Impact Christian Church — Moon Township, PA **Current**

Worship Team Musician (keyboard and vocals) · Worship-Leader Mentor

- Play and sing on the worship team, bringing organ, piano, and contemporary-worship experience to a modern worship setting.
- Mentor emerging worship leaders in musical preparation, flow, and pastoral sensitivity.

Christ Church at Grove Farm — Sewickley, PA **2016 – Present**

Music Ministry · Worship Leader / Director

- Lead music ministry for a congregation founded in 1995 by members of St. Stephen's, with John Guest as its first rector, on the 46-acre Grove Farm property.
- Plan and lead worship music, blending traditional sacred repertoire with contemporary and crossover styles.

Grace Church — Edgeworth, PA **2014 – 2015**

Director of Music

- Directed the parish music program — choral preparation, service playing, and repertoire for the liturgical year — concurrently with a Lead V&V engineering role.

Geneva College — Beaver Falls, PA **9/2010 – 6/2011**

Adjunct Music Faculty

- Taught freshman and sophomore core curriculum — individual (solo) ear training and music theory — the year-long core for future music teachers.
- Taught organ, including organ student Daniel Favand, on the pipe organ in an approximately 300-seat classroom built like a performance stage.
- Led a standing-room-only keyboard class on nine full-size keyboards in the basement of the college's old Victorian music house.
- Student-centered method: students brought two pieces they wanted to learn, then chose one work from the approved list for a recital jury of three music department chairs and the piano chair. Every student who remained in the course earned an A.

St. Stephen's Episcopal Church — Sewickley, PA **1993 – 2009**

Director of Music Ministries (full-time), 1994 – 2009 · Assistant Organist (part-time), 1993 – 1994

- Ran a large, full-range music program for a large Episcopal parish, managing more than 50 people weekly; the rector made the music post full-time in 1994.
- Directed the choral program and served as principal organist and pianist for the full liturgical year under the Book of Common Prayer (1979).
- Served during the parish's growth era under John Guest; built repertoire combining the organ and choral canon with contemporary and crossover styles.

University of Miami — Coral Gables, FL **1988 – 1990**

Graduate Teaching Assistant (full teaching assistantship)

- Taught choir, private piano, and classroom music theory to freshman and sophomore classes while completing the Master of Music in Choral Conducting.

PROFESSIONAL EXPERIENCE — SOFTWARE QUALITY ASSURANCE

Omnicell

Lead Software Quality Assurance Analyst — SQA Level IV

Omnicell is a provider of medication-management and pharmacy-automation solutions — robotics, smart devices, and software — for health systems and pharmacies.

- Built VMware virtual test environments and VM templates for unattended testing, as the SQA engineer on a small team with five developers, led by Derek.
- Worked with HP QTP and a Test Center environment, together with JIRA-style tracking and automation tooling.
- Applied medical-device V&V discipline to test planning, execution, and defect management.

Management Science Associates (MSA) — Pittsburgh, PA **8/2018 – 2/2019**

Software Quality Assurance Analyst, Life Sciences Division · Contract via Pinnakle Technologies

- Web application SQA for the Life Sciences Division, including UI manual testing of EditLife (editlife.msa.com).
- Tested a large-scale, enterprise-level organ-transplant application used across a hospital network, in a HIPAA-compliant environment.
- Worked with Business Administrators to edit and verify requirements and use cases; logged defects in Microsoft Team Foundation Server (TFS).

Software Design Solutions (an Applied Visions company) — Murrysville, PA **5/2017 – 4/2018**

Senior Software Quality Assurance Analyst — Embedded Software & Hardware / IoT (Full-time)

- Embedded with a small development team as the QA/test engineer for bus, train, and ferry transit applications used to run public transit systems in the United States.
- Sole in-house SQA engineer for an internal development team of about 10 engineers working across multiple projects.
- Tested the upgrade of NASA satellite ground-control software (L3 / TTS program — name to confirm), migrated from legacy C++ during the relocation of the system from Arizona to Australia.
- Built an internal VMware ESXi virtualization lab from repurposed servers; created VM templates enabling unattended automated testing alongside manual testing, establishing an early SDLC with multiple live, resourced virtual machines.
- Performed software testing on the F-35 pilot helmet-mounted display, which projects flight and weapons data onto the pilot's visor.
- Performed SQA on embedded software and hardware (IoT) in an Agile environment — functional, negative, stress, integration, and regression testing.
- Created test plans and test cases, with a reporting structure in TestRail, even when requirements were unavailable or incomplete.
- Worked with Ed Kuzemchak, founder of Software Design Solutions.

Celtic Healthcare — Mars, PA **10/2016 – 4/2017**

Software Quality Assurance Engineer · Contract via Oxford Solutions, converted to full-time

- Tested web-based healthcare documentation and medical-coding software in an Agile team.
- UI and functional-integration testing; created and executed SQA tests tracked in JIRA.

Ciber — Pittsburgh, PA **3/2016 – 9/2016**

Senior Software Quality Assurance Engineer / Sr. QA Specialist (Full-time, regular)

- Tested a web-based Institutional Review Board (IRB) system supporting FDA-regulated clinical-trial software and document management.
- UI and functional-integration testing using Seapine TestTrack ALM; wrote and executed SQA test procedures.
- Ran pre-authored and approved SQL queries to capture back-end log data for verification.

Bayer Pharma Radiology & Informatics (formerly Medrad) — Indianola, PA **9/2014 – 2/2016**

Lead Verification and Validation Software Quality Assurance Engineer (Medical Device) · Contract via Premiere Technology Services (PTS), Walton Hills, OH

V&V of CT and MRI contrast-injector systems and control software in an FDA Class I / Class II regulated lab — including the Stellant / Certegra family (FDA Class II), used in hospitals worldwide for pediatric and premature-infant patients, with dual injectors timed between heartbeats.

- Worked on the Certegra MRI injector.
- Built and managed VMware ESXi virtual test environments, Jira, and Microsoft server farms.
- Performed CT and MRI injector V&V under FDA QSR 21 CFR Part 820 and ISO 13485; coordinated V&V plans, protocols, dry runs, test set-up validation, and verification reports.
- Conducted multi-injector physical pressure testing on approximately eight injectors at once, supported by background database scripts, frequently in GE MRI configurations.
- Captured discrepancies (DTRs) and drove them to resolution; performed automation testing with pre-scripted, low-level embedded C++-type code and tested HTML-based UI applications.
- Performed handwritten, hand-signed production release verification on a legacy device.
- Consulted with design teams on the day-in-the-life of the injector across CT, MR, PET, and interventional suites for NPD and LCE projects; guided use cases, task analyses, and workflows.

- Composed and verified controlled documentation before electronic signatures — three master books of roughly 500 pages each.
- Daily scrum with developers and the project lead; "translated" for the SQA team; worked within a team of seven developers and mentored small SQA teams; helped manage the PTS home team (Ohio) with Jim Jennings.
- Tools: IBM Rational DOORS; Enovia PLM and MatrixOne PLM; MKS Integrity Client; JIRA; Confluence.

Philips Respironics — Monroeville, PA **9/2013 – 8/2014**

SQA Verification and Validation Test Engineer · Contract via Canon Recruiting Group, LLC, Santa Clarita, CA

- Design V&V on project teams developing medical devices — mechanical and electro-mechanical systems with embedded software and hardware for patient use.
- Wrote and executed detailed test plans against mechanical, electrical, and software requirements, with traceability across the SDLC.
- Tested legacy and new-hardware product designs; functional-requirements and performance testing.
- Standards: FDA 21 CFR 820 / ISO 13485, IEC 62304, ISO 14971 / IEC 60601. Tools: HP Quality Center, IBM Rational DOORS, Visual SourceSafe, SAP document management.

K&L Gates, LLC — Pittsburgh, PA **9/2011 – 3/2013**

Software Quality Assurance Test Engineer & Application Support · Contract via Regency Group, Ltd., converted to full-time

- Led a start-up QA team for the firm's internal IT organization, implementing system image upgrades: Windows 7 images, remote SCCM deployments, and desktop installs.
- Used VMware vSphere server virtualization in core image testing; created and managed VMware ESXi environments.
- Tracked QA in Microsoft Team Foundation Server (TFS) with Visual Studio and Microsoft Test Manager; Level 3 support for WSUS.

ADDITIONAL PROFESSIONAL EXPERIENCE

Computer Housecall — Pittsburgh, PA **2/1996 – 5/2000**

PC Field Technician · Help Desk — concurrent with my church music post

Penn Financial Services — Pittsburgh, PA

Agent — Long-Term Care, Mutual Funds, Retirement Planning

Public Beach, Oyster Road area — Fairfield, CT **Early career**

Lifeguard and Full-Time Sailing Director

MUSIC LEADERSHIP & PERFORMANCE HISTORY

Formation and early training

- Countertenor in the Episcopal men-and-boys choir in Southport, Connecticut, under Ben Hawks.
- Organ study with Mrs. Larson, Dan Hursey, and Thomas Crawford, with study links to David Craighead; chamber Bach with Igor Kipnis in Greenwich, Connecticut, programs.
- Organ study at Northwestern University with Grigg Fountain and my primary organ instructor, Richard Enright.

Performance & tours

- University of Miami Singers world tours to Australia and to Romania, Yugoslavia, and Austria, performing a half-choral / half show-choir program I wrote.
- Music for a three-city series of large outdoor programs with John Guest, held in parking lots and open-air sites.
- Organ endurance repertoire (e.g., Bach in A minor, Finlandia) with refined stop sense and registration.
- Crossover pianist in Billy Joel, Elton John, and James Taylor repertoire for worship and outreach.

COMPOSITIONS & ARRANGEMENTS

- **Gloria (SATB and organ).** My first fully engraved score; Book of Common Prayer (1979) Rite II text; Edition Peters–style engraving in a Newzik Pro workflow; dedicated in memory of Sir Stephen Cleobury.
- **Magnificat.** Written and performed at the University of Miami; in active revision (lyrics and chords → Moises Pro stems → engraved score).
- **Direction.** Building a body of sacred choral work in the lane of composers such as Joseph M. Martin, aiming for work worthy of an honorary doctorate.

TEACHING & MENTORING

A natural-born teacher of anything to anyone — music, software quality, drawing, and flight.

- **Church music mentoring:** from the men-and-boys choir legacy to the development of adult worship leaders.
- **SQA mentoring:** mentored developer and SQA teams in FDA-regulated environments and led start-up QA teams.
- **Drawing:** "two pages a day" teach-back practice in the Rodgon character-drawing method, building teach-from lesson packs.

EDUCATION & TRAINING

University of Miami — Coral Gables, FL **1996**
Master of Music, Choral Conducting — Full Teaching Assistantship (coursework 1988–1990; thesis and degree 1996)

Northwestern University — Evanston, IL **1984 – 1988**
Bachelor of Music Education — student-teaching placements in three public school districts near Evanston

Community College of Allegheny County (CCAC) — Pittsburgh, PA **1999**
Certificate in Information Technology — IT Networking

CERTIFICATIONS & CREDENTIALS

- Illinois K–12 Teaching Certification (Music Education, with Special Education and music therapy) — Northwestern University.
- Regulated medical-device certifications: FDA 21 CFR 820 / ISO 13485 (quality systems); IEC 62304 (software life cycle); ISO 14971 and IEC 60601 (risk management; electrical safety).

TECHNICAL SKILLS & TOOLS

Test & ALM: HP Quality Center / ALM, HP QTP, Test Center, TestRail, Seapine TestTrack ALM, Microsoft Test Manager, Microsoft Team Foundation Server (TFS), Visual Studio

Requirements & PLM: IBM Rational DOORS, Enovia PLM, MatrixOne PLM, SAP document management, Visual SourceSafe

Tracking & collaboration: JIRA, Confluence, MKS Integrity Client

Data & back end: SQL Server Management Studio, MSSQL, SQLite, Oracle PL/SQL, Chrome REST client

Healthcare & standards: HL7, HIPAA, FDA QSR 21 CFR 820, ISO 13485, IEC 62304, ISO 14971, IEC 60601, ISO 9001

Infrastructure: VMware vSphere / ESXi, SCCM, Active Directory, WSUS, Windows 7 imaging, Microsoft server farms, IT networking

Music technology: Logic Pro (48 kHz), MainStage, Spitfire Audio libraries, Newzik Pro, Moises Pro, AU / VST plug-in concepts

Visual art: Clip Studio Paint, Procreate (iPad), reMarkable

Languages: English; Spanish (bilingual childhood schooling in Bayamón, Puerto Rico)

CREATIVE TECHNOLOGY PROJECTS (CURRENT)

- **Vocalis (in design).** Music-practice and score software combining Moises-class stem separation, a Newzik-class score desk, and an AU / VST plug-in; teacher mode compares a student's take against a reference stem. Product brief and desktop scaffold complete.
- **ManuelOps.** Personal sprint and operations system on MSSQL, with Idea Cards and Monopoly-style status boards; prioritization by planning poker, carried over from Bayer scrum.
- **Rodgon character drawing.** Study of the Rodgon method (Rodrigo Gonzalez) across reMarkable, Clip Studio Paint, and Procreate.

SERVICE & VOLUNTEER WORK

- **YWAM (Youth With A Mission) — Juárez, Mexico.** Taught myself acoustic guitar and led worship bands under pressure for crowds with lines around the corner.
- **Sea Scouts, Ship 84 — Southport, Connecticut.** Member of the Sea Scouts youth maritime program.

REFERENCES

Professional and musical references are available on request.

SOURCES

Public sources supporting selected facts; institution names above link to official sites.

1. Grigg Fountain, Professor of Organ and Church Music, Northwestern (1961–1986) — Northwestern University Archives: https://findingaids.library.northwestern.edu/agents/people/1510
2. Grigg Fountain obituary — The Diapason: https://www.thediapason.com/news/grigg-fountain-dead-97
3. Richard Enright, Professor of Church Music and Organ, Northwestern (35 years, to 1989) — Bienen School of Music: https://www.music.northwestern.edu/news/2016/richard-enright-1923-2016
4. Richard Enright obituary — The Diapason: https://www.thediapason.com/news/richard-enright-dead-93
5. University of Miami choral program history (Lee Kjelson, 1968–1993) — Frost School of Music: https://choral.frost.miami.edu/about-us/choral-program-history/index.html
6. Christ Church at Grove Farm history (founded 1995; John Guest first rector; 46-acre Grove Farm): https://www.ccgf.org/our-history/
7. John Guest biography (St. Stephen's, Sewickley; Christ Church at Grove Farm, 1995): http://www.onevoicemusic.faithweb.com/john1.htm
8. Christianity Today news brief, December 11, 1995 (John Guest and the New Church of Sewickley): https://www.christianitytoday.com/1995/12/news-briefs-december-11-1995-2/
9. Sea Scout Ship 84, Southport, CT — Connecticut Yankee Council: https://www.ctyankee.org/sea-scouts/
10. Pinnakle Technologies — company profile: http://www.pinnakle.net/about.html
11. Omnicell — medication-management and pharmacy-automation solutions (company press release): https://ir.omnicell.com/news/news-details/2018/Omnicell-Opens-Center-of-Excellence-in-Cranberry-Township-Showcasing-Innovation-and-Industry-Leading-Medication-Management-Platform/default.aspx
12. Software Design Solutions — founded 2003 by Ed Kuzemchak; acquired by Applied Visions, September 2016: https://teamsds.com/sensors-midwest-2018/